as it moves through curved spacetime. Not only have these predictions been confirmed to a high level of accuracy, but in head-to-head competition with the predictions of Newton's theory, Einstein's theory consistently matches reality with finer fidelity.

Of equal importance, since general relativity specifies the detailed mechanism by which gravity works, it provides a mathematical framework for determining how fast it transmits its influence. The speed of transmission comes down to the question of how fast the shape of space can change in time. That is, how quickly can warps and ripples—ripples like those on the surface of a pond caused by a plunging pebble—race from place to place through space? Einstein was able to work this out, and the answer he came to was enormously gratifying. He found that warps and ripples—gravity, that is—do not travel from place to place instantaneously, as they do in Newtonian calculations of gravity. Instead, *they travel at exactly the speed of light.* Not a bit faster or slower, fully in keeping with the speed limit set by special relativity. If aliens plucked the moon from its orbit, the tides would recede a second and a half later, at the exact same moment we'd see that the moon had vanished. Where Newton's theory failed, Einstein's general relativity prevailed.

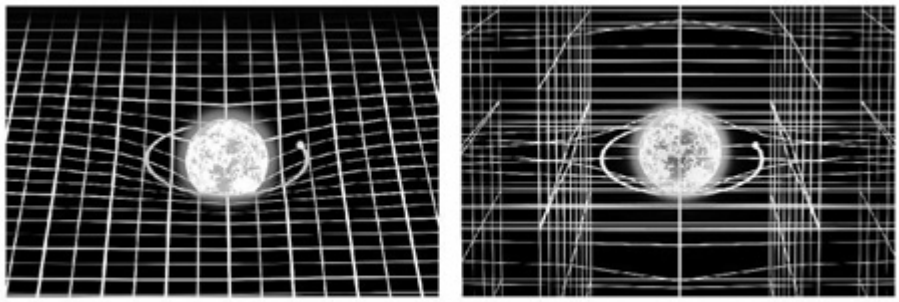


Figure 3.11 The earth stays in orbit around the sun because it follows curves in the spacetime fabric caused by the sun's presence. **(**a **)** 2-d version. **(**b**)** 3-d version.

General Relativity and the Bucket

Beyond giving the world a mathematically elegant, conceptually powerful, and, for the first time, fully consistent theory of gravity, the